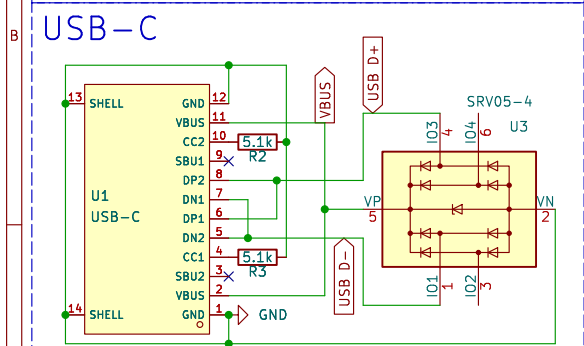


Port-Expander Button Connector

The diagram illustrates a circuit for a Port-Expander Button Connector. The central component is the TCA9555PWR IC (U2), which is configured as a port expander. The circuit includes a +3.3V power supply connected to the VDD pin (pin 24) and a GND connection to the VSS pin (pin 12). The IC's I2C interface is connected to an I2C bus, with the SCL pin (pin 22) connected to the I2C_SCL pin (pin 1) and the SDA pin (pin 23) connected to the I2C_SDA pin (pin 2). A 10kΩ pull-up resistor (R20) is connected to the SCL line, and another 10kΩ pull-up resistor (R21) is connected to the SDA line. The INT pin (pin 1) is connected to a WAKEUP signal. The IC's output pins (pins 4 through 20) are connected to a button connector (J1) and a button (BTN1-4). The button connector pins are labeled as follows: 4 (I/O0_0), 5 (I/O0_1), 6 (I/O0_2), 7 (I/O0_3), 8 (I/O0_4), 9 (I/O0_5), 10 (I/O0_6), 11 (I/O0_7), 13 (I/O1_0), 14 (I/O1_1), 15 (I/O1_2), 16 (I/O1_3), 17 (I/O1_4), 18 (I/O1_5), 19 (I/O1_6), and 20 (I/O1_7). The button connector is also connected to a button (BTN1-4) and a button (BTN5-8). The button (BTN1-4) is connected to the I/O0_0 pin (pin 4) and the I/O0_1 pin (pin 5). The button (BTN5-8) is connected to the I/O0_2 pin (pin 6) and the I/O0_3 pin (pin 7). The button (BTN1-4) is also connected to the I/O0_4 pin (pin 8) and the I/O0_5 pin (pin 9). The button (BTN5-8) is also connected to the I/O0_6 pin (pin 10) and the I/O0_7 pin (pin 11). The button (BTN1-4) is also connected to the I/O0_8 pin (pin 12) and the I/O0_9 pin (pin 13). The button (BTN5-8) is also connected to the I/O0_10 pin (pin 14) and the I/O0_11 pin (pin 15). The button (BTN1-4) is also connected to the I/O0_12 pin (pin 16) and the I/O0_13 pin (pin 17). The button (BTN5-8) is also connected to the I/O0_14 pin (pin 18) and the I/O0_15 pin (pin 19). The button (BTN1-4) is also connected to the I/O0_16 pin (pin 20) and the I/O0_17 pin (pin 21). The button (BTN5-8) is also connected to the I/O0_18 pin (pin 22) and the I/O0_19 pin (pin 23). The button (BTN1-4) is also connected to the I/O0_20 pin (pin 24) and the I/O0_21 pin (pin 25). The button (BTN5-8) is also connected to the I/O0_22 pin (pin 26) and the I/O0_23 pin (pin 27). The button (BTN1-4) is also connected to the I/O0_24 pin (pin 28) and the I/O0_25 pin (pin 29). The button (BTN5-8) is also connected to the I/O0_26 pin (pin 30) and the I/O0_27 pin (pin 31). The button (BTN1-4) is also connected to the I/O0_28 pin (pin 32) and the I/O0_29 pin (pin 33). The button (BTN5-8) is also connected to the I/O0_30 pin (pin 34) and the I/O0_31 pin (pin 35). The button (BTN1-4) is also connected to the I/O0_32 pin (pin 36) and the I/O0_33 pin (pin 37). The button (BTN5-8) is also connected to the I/O0_34 pin (pin 38) and the I/O0_35 pin (pin 39). The button (BTN1-4) is also connected to the I/O0_36 pin (pin 40) and the I/O0_37 pin (pin 41). The button (BTN5-8) is also connected to the I/O0_38 pin (pin 42) and the I/O0_39 pin (pin 43). The button (BTN1-4) is also connected to the I/O0_40 pin (pin 44) and the I/O0_41 pin (pin 45). The button (BTN5-8) is also connected to the I/O0_42 pin (pin 46) and the I/O0_43 pin (pin 47). The button (BTN1-4) is also connected to the I/O0_44 pin (pin 48) and the I/O0_45 pin (pin 49). The button (BTN5-8) is also connected to the I/O0_46 pin (pin 50) and the I/O0_47 pin (pin 51). The button (BTN1-4) is also connected to the I/O0_48 pin (pin 52) and the I/O0_49 pin (pin 53). The button (BTN5-8) is also connected to the I/O0_50 pin (pin 54) and the I/O0_51 pin (pin 55). The button (BTN1-4) is also connected to the I/O0_52 pin (pin 56) and the I/O0_53 pin (pin 57). The button (BTN5-8) is also connected to the I/O0_54 pin (pin 58) and the I/O0_55 pin (pin 59). The button (BTN1-4) is also connected to the I/O0_56 pin (pin 60) and the I/O0_57 pin (pin 61). The button (BTN5-8) is also connected to the I/O0_58 pin (pin 62) and the I/O0_59 pin (pin 63). The button (BTN1-4) is also connected to the I/O0_60 pin (pin 64) and the I/O0_61 pin (pin 65). The button (BTN5-8) is also connected to the I/O0_62 pin (pin 66) and the I/O0_63 pin (pin 67). The button (BTN1-4) is also connected to the I/O0_64 pin (pin 68) and the I/O0_65 pin (pin 69). The button (BTN5-8) is also connected to the I/O0_66 pin (pin 70) and the I/O0_67 pin (pin 71). The button (BTN1-4) is also connected to the I/O0_68 pin (pin 72) and the I/O0_69 pin (pin 73). The button (BTN5-8) is also connected to the I/O0_70 pin (pin 74) and the I/O0_71 pin (pin 75). The button (BTN1-4) is also connected to the I/O0_72 pin (pin 76) and the I/O0_73 pin (pin 77). The button (BTN5-8) is also connected to the I/O0_74 pin (pin 78) and the I/O0_75 pin (pin 79). The button (BTN1-4) is also connected to the I/O0_76 pin (pin 80) and the I/O0_77 pin (pin 81). The button (BTN5-8) is also connected to the I/O0_78 pin (pin 82) and the I/O0_79 pin (pin 83). The button (BTN1-4) is also connected to the I/O0_80 pin (pin 84) and the I/O0_81 pin (pin 85). The button (BTN5-8) is also connected to the I/O0_82 pin (pin 86) and the I/O0_83 pin (pin 87). The button (BTN1-4) is also connected to the I/O0_84 pin (pin 88) and the I/O0_85 pin (pin 89). The button (BTN5-8) is also connected to the I/O0_86 pin (pin 90) and the I/O0_87 pin (pin 91). The button (BTN1-4) is also connected to the I/O0_88 pin (pin 92) and the I/O0_89 pin (pin 93). The button (BTN5-8) is also connected to the I/O0_90 pin (pin 94) and the I/O0_91 pin (pin 95). The button (BTN1-4) is also connected to the I/O0_92 pin (pin 96) and the I/O0_93 pin (pin 97). The button (BTN5-8) is also connected to the I/O0_94 pin (pin 98) and the I/O0_95 pin (pin 99). The button (BTN1-4) is also connected to the I/O0_96 pin (pin 100) and the I/O0_97 pin (pin 101). The button (BTN5-8) is also connected to the I/O0_98 pin (pin 102) and the I/O0_99 pin (pin 103). The button (BTN1-4) is also connected to the I/O0_100 pin (pin 104) and the I/O0_101 pin (pin 105). The button (BTN5-8) is also connected to the I/O0_102 pin (pin 106) and the I/O0_103 pin (pin 107). The button (BTN1-4) is also connected to the I/O0_104 pin (pin 108) and the I/O0_105 pin (pin 109). The button (BTN5-8) is also connected to the I/O0_106 pin (pin 110) and the I/O0_107 pin (pin 111). The button (BTN1-4) is also connected to the I/O0_108 pin (pin 112) and the I/O0_109 pin (pin 113). The button (BTN5-8) is also connected to the I/O0_110 pin (pin 114) and the I/O0_111 pin (pin 115). The button (BTN1-4) is also connected to the I/O0_112 pin (pin 116) and the I/O0_113 pin (pin 117). The button (BTN5-8) is also connected to the I/O0_114 pin (pin 118) and the I/O0_115 pin (pin 119). The button (BTN1-4) is also connected to the I/O0_116 pin (pin 120) and the I/O0_117 pin (pin 121). The button (BTN5-8) is also connected to the I/O0_118 pin (pin 122) and the I/O0_119 pin (pin 123). The button (BTN1-4) is also connected to the I/O0_120 pin (pin 124) and the I/O0_121 pin (pin 125). The button (BTN5-8) is also connected to the I/O0_122 pin (pin 126) and the I/O0_123 pin (pin 127). The button (BTN1-4) is also connected to the I/O0_124 pin (pin 128) and the I/O0_125 pin (pin 129). The button (BTN5-8) is also connected to the I/O0_126 pin (pin 130) and the I/O0_127 pin (pin 131). The button (BTN1-4) is also connected to the I/O0_128 pin (pin 132) and the I/O0_129 pin (pin 133). The button (BTN5-8) is also connected to the I/O0_130 pin (pin 134) and the I/O0_131 pin (pin 135). The button (BTN1-4) is also connected to the I/O0_132 pin (pin 136) and the I/O0_133 pin (pin 137). The button (BTN5-8) is also connected to the I/O0_134 pin (pin 138) and the I/O0_135 pin (pin 139). The button (BTN1-4) is also connected to the I/O0_136 pin (pin 140) and the I/O0_137 pin (pin 141). The button (BTN5-8) is also connected to the I/O0_138 pin (pin 142) and the I/O0_139 pin (pin 143). The button (BTN1-4) is also connected to the I/O0_140 pin (pin 144) and the I/O0_141 pin (pin 145). The button (BTN5-8) is also connected to the I/O0_142 pin (pin 146) and the I/O0_143 pin (pin 147). The button (BTN1-4) is also connected to the I/O0_144 pin (pin 148) and the I/O0_145 pin (pin 149). The button (BTN5-8) is also connected to the I/O0_146 pin (pin 150) and the I/O0_147 pin (pin 151). The button (BTN1-4) is also connected to the I/O0_148 pin (pin 152) and the I/O0_149 pin (pin 153). The button (BTN5-8) is also connected to the I/O0_150 pin (pin 154) and the I/O0_151 pin (pin 155). The button (BTN1-4) is also connected to the I/O0_152 pin (pin 156) and the I/O0_153 pin (pin 157). The button (BTN5-8) is also connected to the I/O0_154 pin (pin 158) and the I/O0_155 pin (pin 159). The button (BTN1-4) is also connected to the I/O0_156 pin (pin 160) and the I/O0_157 pin (pin 161). The button (BTN5-8) is also connected to the I/O0_158 pin (pin 162) and the I/O0_159 pin (pin 163). The button (BTN1-4) is also connected to the I/O0_160 pin (pin 164) and the I/O0_161 pin (pin 165). The button (BTN5-8) is also connected to the I/O0_162 pin (pin 166) and the I/O0_163 pin (pin 167). The button (BTN1-4) is also connected to the I/O0_164 pin (pin 168) and the I/O0_165 pin (pin 169). The button (BTN5-8) is also connected to the I/O0_166 pin (pin 170) and the I/O0_167 pin (pin 171). The button (BTN1-4) is also connected to the I/O0_168 pin (pin 172) and the I/O0_169 pin (pin 173). The button (BTN5-8) is also connected to the I



The schematic diagram illustrates the power supply and pin connections for the ESP32-WROVER-E module. The module is represented by a yellow rectangle with various pins and internal components labeled.

Power Supply and Decoupling:

- A +3.3V supply is connected to the EN/CHIP_PU pin (pin 3) via a 10k resistor (R6).
- Decoupling capacitors are shown: 100uF capacitors (C16, C15) and a 10uF capacitor (C5) connected to GND.
- The 3V3 pin (pin 2) is also connected to GND.
- A GND_THERMAL pin (pin 9) is connected to GND.

Pin Connections (Left Side):

- ESP32_EN (pin 3) is connected to +3.3V via R6.
- RFID_NSS (pin 8) and RFID_RST (pin 9) are connected to a common line.
- LED_DI (pin 14) and RFID_BUSY (pin 13) are connected to a common line.
- MMC_CLK (pin 23) and MMC_MOSI (pin 22) are connected to a common line.
- ROTENC_BTN (pin 6) is connected to a common line.
- WAKEUP (pin 7) is connected to a common line.
- ROTENC_DAT (pin 4) and ROTENC_CLK (pin 5) are connected to a common line.

Module Labels and Pins:

- EN/CHIP_PU (pin 3)
- 32K_XP/GPIO32/ADC1_CH4 (pin 8)
- 32K_XN/GPIO33/ADC1_CH5 (pin 9)
- GPIO00/BOOT0 (pin 10)
- GPIO10/BOOT1 (pin 11)
- MTDI/GPIO12/ADC2_CH5 (pin 14)
- MTCK/GPIO13/ADC2_CH4 (pin 13)
- MTMS/GPIO14/ADC2_CH6 (pin 23)
- MTDO/GPIO15/ADC2_CH3 (pin 22)
- GPI00/BOOT0 (pin 10)
- ESP32-WROVER-E (U4)
- ADC2_CH0 (pin 11)
- ADC2_CH1 (pin 12)
- GPI034/ADC1_CH6 (pin 6)
- GPI035/ADC1_CH7 (pin 7)
- SENSOR_VP/GPIO36/ADC1_CH0 (pin 4)
- SENSOR_VN/GPIO39/ADC1_CH3 (pin 5)
- DAC1/ADC2_CH0 (pin 10)
- DAC2/ADC2_CH1 (pin 11)
- ADC2_CH2 (pin 12)
- ADC2_CH3 (pin 13)

Power Supply Section (Bottom):

- Battery BT1 and Solar BT2 are connected to the power supply section.
- VBAT and SOLAR pins are connected to the power supply section.
- GND is connected to the power supply section.
- The power supply section is labeled "Battery Management" and "File: LiPoCharger.kicad_sch".
- VBUS (pin 1) is connected to Vin.
- 3.3VD and SWITCH_3.3VD are connected to +3.3V.
- IRQD (pin 14) is connected to BAT_IRQ.
- SDA (pin 21) is connected to I2C_SDA.
- SCK (pin 22) is connected to I2C_SCL.
- AGND (pin 15) is connected to GND.

Diagram of a 10-pin header with labels:

- 1: GND
- 2: +3.3V
- 3: +3.3V
- 4: GPIO_0
- 5: GPIO_4
- 6: GPIO_5
- 7: I2C_SDA
- 8: I2C_SCL
- 9: UART_TX
- 10: UART_RX

Rotary Encoder

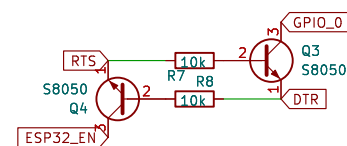
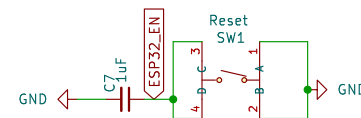
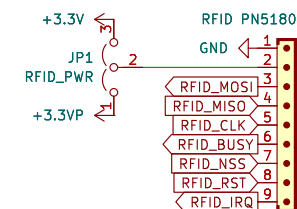
Diagram of a 5-pin rotary encoder connector (J4). The pins are numbered 1 to 5. The connections are:

- Pin 1: GND
- Pin 2: +3.3V
- Pin 3: ROTENC_DAT
- Pin 4: ROTENC_CLK
- Pin 5: ROTENC_BTN

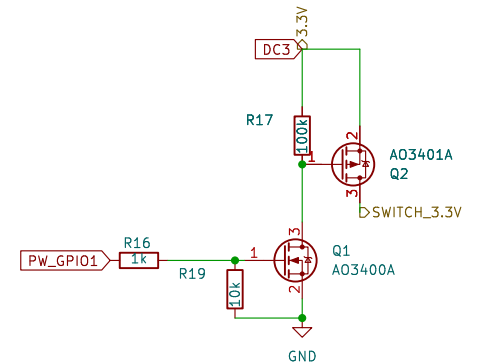
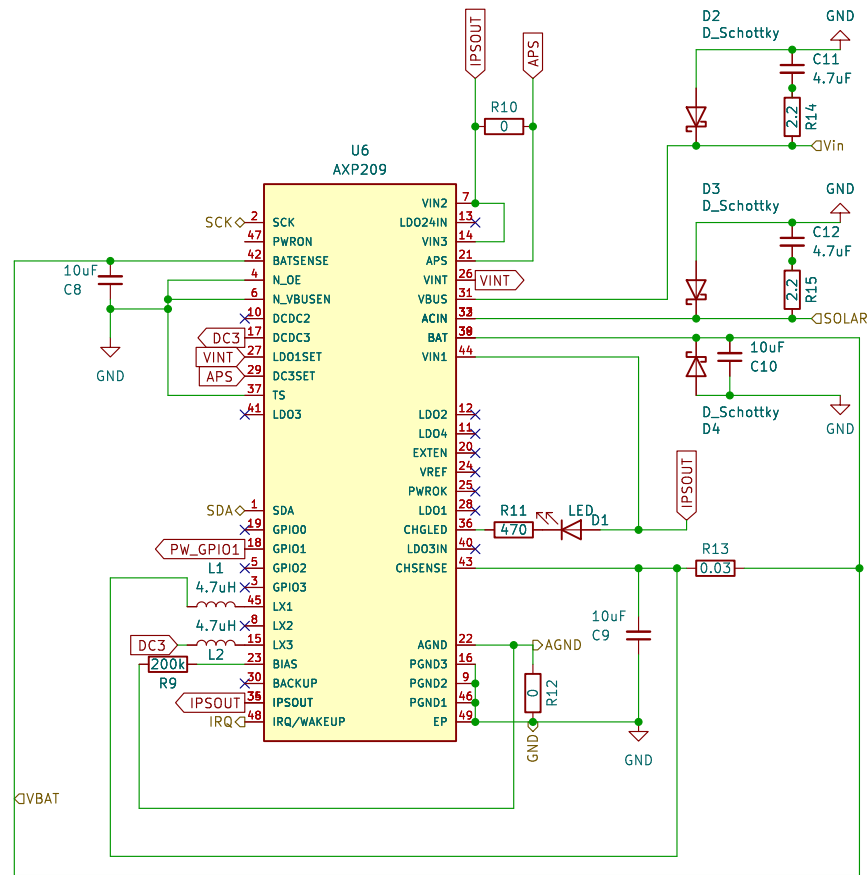
Neopixel

The diagram illustrates the wiring for a Neopixel LED. A +3.3VP power source is connected to the VCC pin (pin 1) of the J5 LED strip. A 10uF capacitor is connected between the VCC pin and GND. The GND pin (pin 2) is connected to a common ground. The data pin (pin 3) is connected to the LED strip. The LED strip is labeled 'LED-DI' and 'J5 LED'.

Auto-Reset



Sheet: /	
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Title: Quino Complete	
Size: A4	Rev: 0.0.2
Date: 2024-12-29	
KiCad E.D.A. 8.0.7	Id: 1/3



Sheet: /Battery Management/
File: LiPoCharger.kicad_sch

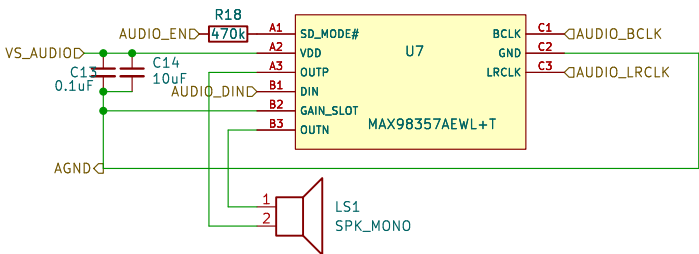
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Size: A4 Date: 2024-12-29

KiCad E.D.A. 8.0.7

Rev: 0.0.2

Id: 2/3



Sheet: /Audio/ File: Audio.kicad_sch		
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Size: A4	Date: 2024-12-29	Rev: 0.0.2
KiCad E.D.A. 8.0.7		Id: 3/3